Mini Project

Textile Production, Cost & Quality Management System

Design and develop a menu-driven Textile Production, Cost and Quality Management System using C programming language. The system should perform basic calculations related to yarn, fabric production, costing, quality inspection, and inventory management.

1. Yarn Count Calculator

The system should accept yarn length and weight and calculate the selected yarn count.

- **Tex:** Tex = Weight (g) / Length (km)
- **Denier:** Denier = [Weight (g) / Length (m)] × 9000
- **Metric Count (Nm):** Nm = Length (m) / Weight (g)

2. Fabric GSM Calculator

The system should calculate the Grams per Square Metre (GSM) of fabric.

- **GSM:** GSM = Weight of fabric (g) / Area (m²)
- **Area when dimensions are in cm:** Area (m²) = [Length (cm) × Width (cm)] / 10000

3. Fabric Production Calculator

The system should estimate fabric production using machine/loom parameters.

- **Basic production:** Production = Machine Speed × Working Time × Efficiency
- **Efficiency-adjusted production:** Effective Production = Machine Speed × Working Time × (Efficiency / 100)
- **Multiple machines:** Total Production = Production per Machine × Number of Machines

4. Fabric Production Cost Calculator

The system should calculate total production cost using yarn, processing, labour and other expenses.

- **Total cost:** Total Cost = Yarn + Weaving + Dyeing + Finishing + Labour + Other Costs
- **Cost per metre:** Cost per metre = Total Cost / Total Fabric Production (m)
- **Profit:** Profit = Cost per metre × (Profit % / 100)

- **Selling price:** Selling Price = Cost per metre + Profit

5. Fabric Quality & Defect Analysis

Record defects such as holes, stains, broken ends, broken picks, slubs, and other defects.

- **Total defects:** Total Defects = D1 + D2 + D3 + ... + Dn
- **Defect percentage:** Defect % = (Total Defects / Total Fabric Inspection Units) × 100

6. Textile Inventory Management

Maintain basic stock records for yarn, fabric, dyes, chemicals and accessories.

- **Low-stock condition:** If Current Stock ≤ Reorder Level → Low Stock – Reorder Required

Example for GSM:

Fabric weight = 50 g; Length = 50 cm; Width = 40 cm. Area = (50 × 40) / 10000 = 0.20 m². GSM = 50 / 0.20 = 250 GSM.

Quality Classification

Defect %	Quality Grade
< 2%	Excellent
2% – <5%	Good
5% – <10%	Average
≥10%	Poor

These limits are sample project criteria and should not be treated as industry standards.

Inventory Management Features

1. Add a new item
2. Display all items
3. Search an item
4. Update quantity
5. Issue/remove quantity
6. Check low-stock items

7. Consolidated Textile Report

- Yarn count
- Fabric GSM
- Total production

- Cost per metre
- Selling price per metre
- Total defects
- Defect percentage
- Quality grade
- Most common defect
- Current inventory status
- Low-stock items

Suggested Main Menu

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 TEXTILE PRODUCTION & QUALITY MANAGEMENT
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1. Yarn Count Calculator
2. Fabric GSM Calculator
3. Production Calculator
4. Cost Calculator
5. Quality & Defect Analysis
6. Inventory Management
7. Generate Final Report
8. Exit

Enter your choice:
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Important Instructions

- Students may use the formulas provided in this project brief.
- Students should clearly mention the formula, input parameters, units, and assumptions used in their program.
- The emphasis will be on correct implementation of C programming concepts, not advanced textile theory.
- Students should be able to explain their code during the viva.
- Wherever necessary, students should perform input validation, such as preventing division by zero or accepting negative quantities.
- Students should divide the program into appropriate functions rather than writing the entire program inside main().